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Experiment data-driven modeling of tokamak discharge in EAST

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Abstract. A neural network model of tokamak discharge is developed based on the experimental dataset of a superconducting long-pulse tokamak (EAST) campaign 2016-2018. The purpose is to reproduce the response of diagnostic signals to actuators signals without introducing additional physical models. In the present work, the discharge curves of electron density n_e, stored energy W_{mhd}, and loop voltage V_{loop} were reproduced from a series of actuators signals. For n_e and W_{mhd}, the average similarity between the modeling results and the experimental data achieve 89% and 97%, respectively. The promising results demonstrate that the data-driven methodology provides an alternative to the physical-driven methodology for tokamak discharge modeling. The method presented in the manuscript has the potential of being used for validating tokamak's experimental proposals, which could advance and optimize experimental planning and validation.

Keywords: tokamak, discharge modeling, machine learning Submitted to: *Nucl. Fusion*

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1. Introduction

A reliable model of the tokamak is critical to magnetic confinement fusion experimental research. It is used to check the feasibility of a pulse, interpret experimental data, validate the theoretical model, and develop control technology.

The conventional physical-driven modeling tools come from empirical models or derivations based on first principles, the so-called "Integrated Modeling" [1]. "Integrated Modeling" is a suite of module codes that addresses the different physical processes in the tokamak, i.e. core transport, equilibrium, stability, boundary physics, heating, fueling, and current drive. Typical workflows are ETS [1], PTRANSP [2], TSC [3], CRONOS [4], JINTRAC [5], METIS [6], ASTRA [7], TOPICS [8], etc. The reliability of the the first-principles model depends on the completeness of the physical processes involved. In the past few decades, sophisticated physical modules have been developed and integrated into these codes for more realistic modeling results. Typical workflow for a full discharge modeling on tokamak is using sophisticated modules that integrate many physical processes [1, 9]. Due to the nonlinear, multi-scale, multi-physics characteristics of tokamak, high-fidelity simulation of the whole tokamak discharge process is still a great scientific challenge [10].

Increasingly, researchers are turning to data-driven approaches. The history can be traced back to the use of machine learning for disruption prediction since the 1990s, i.e. ADITYA [11, 12], Alcator C-Mod [13–15], EAST [15–17], DIII-D [13, 15, 16, 18–28], JET [16, 22, 28–36], ASDEX-Upgrade[30, 37–39], JT-60U [40–42], HL-2A [43, 44], NSTX [45], and J-TEXT [46, 47]. We note that applications of neural networks in the fusion community are increasing rapidly, and examples are simulation acceleration [48–50], plasma tomography [36], radiated power estimation [51], identifying instabilities [52], estimating neutral beam effects [53], classifying confinement regimes [54], determination of scaling laws [55, 56], filament detection on MAST-U [57], electron temperature profile estimation via SXR with Thomson scattering [58], coil current prediction with the heat load pattern in W7-X [59], equilibrium reconstruction [58, 60–64], and equilibrium solver [65]. Additionally, machine learning methods have been tested for control purposes, in some cases to accelerate the computation [66–70]. The Bishop [66] work is the first use of neural networks for the real-time feedback control in a tokamak fusion experiment.

Physics-driven approaches reconstruct physical high-dimensional reality from the bottom-up and then reduce them to the low-dimensional model. Alternatively, data-driven approaches discover the relationships between low-dimensional quantities from a large amount of data and then construct approximate models of the nonlinear dynamical system. When focusing only on the evolution of low-dimensional macroscopic features of complex dynamic systems, data-driven approaches can directly construct models without explicitly considering the physical processes behind the data. In practical applications, actuators signals and diagnostic signals of tokamak usually appear as temporal sequence of low-dimensional data, most of which are zero-dimensional or one-dimensional profiles and rarely two-dimensional distributions. If we consider tokamak as a black box, these signals can be considered as inputs and outputs of a dynamic system. Discharge modeling aims to model the connection between the input and output signals. This can be described as the conversion from one kind of time series data to another. Many deep learning architectures have been created and successfully applied in sequence learning problems [27, 36, 71–73] in areas of time series analysis or natural language processing. At present, machine learning work in the fusion community rarely studies the problem of plasma discharge modeling but understanding the evolution of the plasma state of the tokamak and its response to external control during the discharge process is important. This evolution of the plasma state is useful for the plasma control system and validation of the

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experimental proposals of tokamak.

In the present work, a neural network model is trained with the temporal sequences of actuators signals and diagnostic signals for a large dataset of EAST discharges [74–76]. It can reproduce the response of the main diagnostic signals (such as the electron density n_e, the stored energy W_{mhd}, and the loop voltage V_{loop}.) to actuators signals (such as Lower Hybrid Wave (LHW), Neutral Beam Injection (NBI), etc.) during the whole discharge process and predict their time evolution curves.

The rest of this paper consists of five parts. Section 2 provides descriptions of data preprocessing and data selection criteria. Section 3 shows the model details of this work. The detailed model training process can be found in section 4. Then an in-depth analysis of the model results is put forward in section 5. Finally, a brief conclusion is made in section 6.

2. Dataset

EAST's data system stores more than 3000 channels of raw acquisition signals and thousands of processed physical analysis data [77], which record the entire process of tokamak discharge. These data can be divided into three categories: configuration parameters, actuators signals, and diagnostic signals. The configuration parameters describe constants related to device construction, such as the shape of the vacuum chamber, the position of the poloidal magnetic field (PF) coils, etc. The actuators signals are the external constraints actively applied to the magnetic field coil and auxiliary heating systems, such as the current of PF coils, the power of Lower Hybrid Wave (LHW), etc. The diagnostic signals are the physics information passively measured from the plasma, such as electron density n_e, loop voltage V_{loop}, etc. The configuration parameters will not change during the experimental campaign, so there is no need to consider them unless a cross-device model is built. The discharge modeling can be essentially reduced to a process of mapping actuators (input) signals to diagnostic (output) signals.

In the present work, three signals that can represent the key characteristics of discharge are selected as outputs, which are stored energy W_{mhd}, electron density n_e and loop voltage V_{loop}. The input signal should include all signals that may affect the output. In this paper, ten types of signals are selected as inputs, i.e. plasma current I_p, central toroidal magnetic field B_{t0}, current of PF coils, power of LHW, the raw signals data of neutral Beam Injection (NBI), the raw signals data of Ion Cyclotron Resonance Heating (ICRH), the raw signals data of Electron Cyclotron Resonance Heating/Current Drive (ECRH/ECCD), the raw signals data of Gas Puffing System (GPS), the raw signals data of Supersonic Molecular Beam Injection (SMBI), the raw signal data of Pellet Injection System (PIS). In principle, these signals can be designed in the experimental proposal stage. Detailed information about input and output signals are listed in table 1. Some of these signals are processed signals with a clear physical meaning, and others are unprocessed raw acquisition signals. Since some signals of the actual EAST experimental diagnostic system are not processed, in some cases we directly selected the unprocessed raw acquisition signals. As long as the input signals contains information to determine the output, whether it is a processed physical signal will not affect the modeling result.

Tokamak discharge evolution is a complex nonlinear process, and there is no simple way to determine the connection between the actuators signals and the diagnostic signals. Therefore, the input dataset covers most of the actuators signals so the feature selection of input data is minimized. This input dataset helps prevent the loss of key physical processes in the model. We included all signals related to the evolution of tokamak discharge. Including all signals related to the tokamak

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Table 1. The list of signals. The "raw signal" means the original electrical signal and these could be converted to the signals with physical meaning.

Signals	Physics meanings	Unit	Number of Channels	Sampling rate	Meaning of Channels
Output Signals			3		
n_e	Electron density	$10^{19}m^{-3}$	1	$1kHz$	One chord averaged n_e
V_{loop}	Loop voltage	V	1	$1kHz$	Loop voltage
W_{mhd}	Plasma stored energy	J	1	$20Hz$	Stored energy
Input Signals			65		
I_p	Plasma current	A	2	$1kHz$	Actual and reference plasma current
PF	Current of Poloidal field (PF) coils	A	14	$1kHz$	PF current of 14 Rogowski loops
B_{t0}	Toroidal magenetic field	T	1	$1kHz$	Toroidal field at magnetic axis
LHW	Power of Lower Hybrid Wave Current Drive and Heating System	kW	4	$20kHz$	2.45 GHz LHW, and 4.6 GHz LHW
NBI	Neutral Beam Injection System	Raw signal	8	$5kHz$	Acceleration voltage and beam current, of No. 1-2 left/right ion source.
ICRH	Ion Cyclotron Resonance Heating System	Raw signal	16	$5kHz$	Output of detector for rejected power of No. 1-16 transmitter
ECRH/ ECCD	Electron Cyclotron Resonance Heating/Current Drive System	Raw signal	4	$50kHz$	Output of detector for injected power measurement No. 1-4 gyrotron
GPS	Gas Puffing System	Raw signal	12	$10kHz$	Horizontal ports J, K, D, B; Upper port O; Lower ports O, C, H
SMBI	Supersonic Molecular Beam Injection	Raw signal	3	$10kHz$	3 ports of SMBI
PIS	Pellet Injection System	Raw signal	1	$10kHz$	1 injection line for Pellet Injection

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discharge evolution will introduce some redundant information. For instance, the 14 PF currents are highly correlated and some other signals also have potential correlations. The redundancy in the training dataset will increase the machine learning model training time and reduce modeling accuracy. How to filter input signals is an important issue that needs to be explored in the next step.

There are significant differences in the sampling rate of raw signals R^i_{raw}, where i is the index of signals. The input and output signal data sets need to be resampled at a common sampling rate R_c to ensure that the data points of different signals are aligned at the same time. If $R^i_{raw} < R_c$, the raw signal data needs to be interpolated to complement the time series. If $R^i_{raw} > R_c$, the raw signal data needs to be down-sampled into same sampling rate with R_c by simple moving average (SMA). In the present work, the simple moving average (SMA) calculates the average of a specific time range ($1/R_c$) of values. The SMA used the information from the later time, but the purpose of our work is to create an offline modeling. In an offline modeling, the later time information is already known. So the later time information will not affect the use of the model. From another perspective, experimenter does not design high-frequency fluctuations in the experimental proposal stage. Filtering out high-frequency fluctuations is more appropriate for the application of the model in experimental program validation.

The resampling rate R_c depends on the time resolution of the output signal, which refers to the time resolution of the physical process of interest rather than the sampling rate of the raw experiment signal. The accuracy of the reproduction of the physical process determines the quality of the modeling. The size of the data set determines the length of the model training time and the efficiency of modeling. A lower resampling rate means lower time resolution and poor modeling quality. However, a higher resampling rate means greater computing resource requirements and lower modeling efficiency.

The normal discharge waveform of the tokamak can be divided into three phases, ramp-up, flat-top, and ramp-down (see figure 1). Most signals rise slowly during the ramp-up phase, remain stable during the flat-top phase, and slowly decrease during the ramp-down phase. The time scale of the ramp-up and ramp-down phases are similar, and the flat-top phase is much longer than the former two. The output signals show different time characteristics in these three phases. The signal waveforms of n_e and W_{mhd} remain smooth in all three phases, which can be accurately reproduced with a uniform resampling rate $R^{n_e}_c = 1kHz$. However, the waveform of V_{loop} varies greatly and frequently in the ramp-up and ramp-down phases, see figure 1. In order to ensure the quality of modeling, the resampling rate of V_{loop} is increased in these two phases

$$R^{V_{loop}}_c = \begin{cases} 1kHz, & \text{flat-top}, \\ 10kHz, & \text{ramp-up or ramp-down} \end{cases},$$

which is an adaptive piece-wise function. The purpose of using a non-uniform adaptive resampling function is to balance the quality and efficiency of modeling.

Three signals are selected as outputs, and 65 channels are selected as inputs, and then resampled according to the characteristics of the output signal to align the time points. In the next step, machine learning will be performed on these data.

3. Machine learning architecture

The data of the tokamak diagnostic system are all temporal sequences, and different signal data have different characteristics. According to the temporal characteristics of the data of tokamak diagnostic

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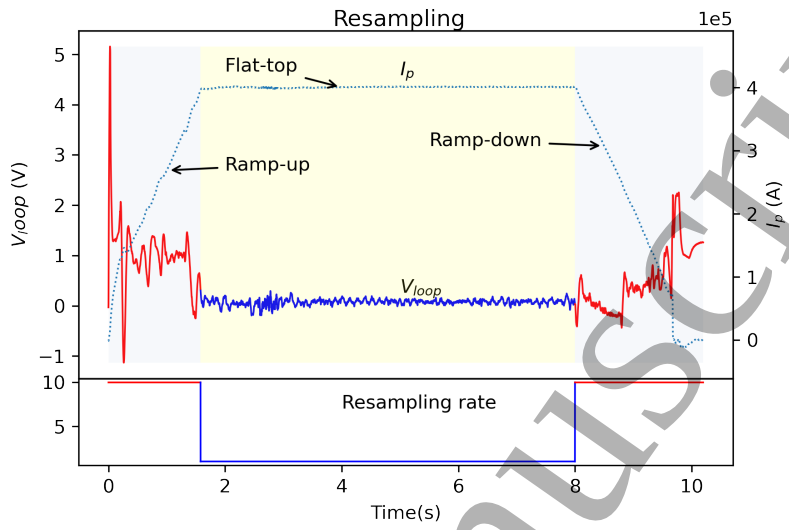

Figure 1. Schematic diagram of adaptive resampling. The higher resampling rate is used for segments that are of great interest to physicists or vary greatly and frequently. The bottom figure is an adaptive piece-wise function of $R_c^{V_{loop}}$. We used the different resampling rates among ramp-up, ramp-down, and flat-top.

system, the sequence to sequence (seq2seq) model [78] was chosen as the machine learning model for tokamak discharge modeling. Two methods of uniform resampling and adaptive resampling are adopted for different data characteristic.

Natural language processing (NLP) is more similar to the discharge modeling process than classification. NLP converts a natural language sequence into another natural language sequence. It is an important branch of machine learning, and the main algorithms are routinely employed recurrent neural network (RNN), gated recurrent unit (GRU) and long short-term memory (LSTM), etc.

The encoder-decoder architecture [78] is a useful architecture in NLP, the architecture is partitioned into two parts, the encoder and the decoder. The encoder's role is to encode the inputs into the state, which often contains several tensors. Then the state is passed into the decoder to generate the outputs. In the present work, LSTM was chosen as a fundamental component of the model. And stacked LSTM as an encoder-decoder machine learning model.

As shown in figure 2, the machine learning model architecture used in the present work is based on the sequence to sequence model (seq2seq) [78]. The first two LSTM layers (LSTM_0 and LSTM_1 in figure 2) and Dropout_0 can be considered as the encoder, and the last two LSTM layers (LSTM_2 and LSTM_3 in figure 2) and Dropout_1 can be regarded as the decoder. In the present work, the encoder learns the high-level representation (*cannot be displayed directly*) of input signals (table 1 input signals). The last hidden state of the encoder is used to *initialize* the hidden state of the decoder. The decoder plays the role of decoding the information of the encoder. By creating a high-level representation of the input features the encoder increases the dimensionality of the input sequence while the decoder learns how to reduce this back to the originally fed time traces. Encoder-decoder is built as an end-to-end model, it can learn sequence information directly

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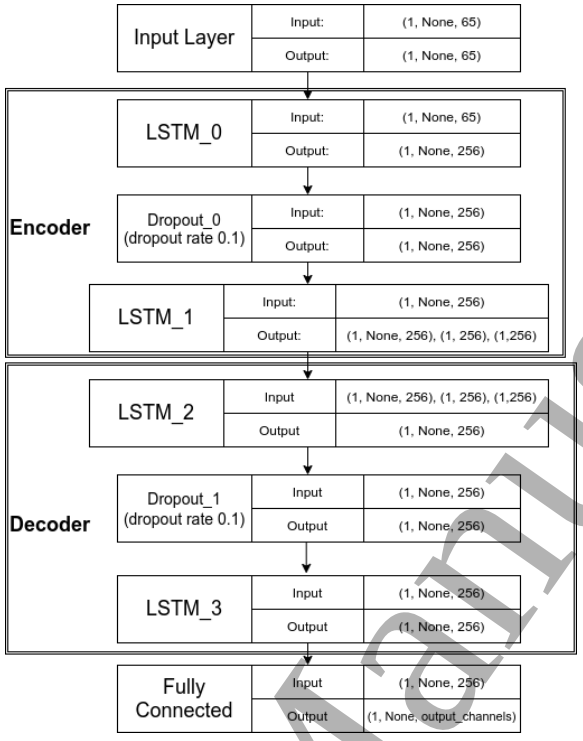

Figure 2. Architecture of our model. The input and output style is (batch size, sequence length, feature size) [79], such as (1, None, 65), where "None" represents different lengths of sequence because of different discharge shot duration time, "1" represents the input batch size is one, "65" means the feature size is 65. "Output_channels" is the number of output sequences.

without manually extracting features.

In terms of components, the main component of our architecture is long short-term memory (LSTM) [80], because the LSTM can use trainable parameters to balance long-term and short-term dependencies. This feature is suitable for tokamak sequence data. Tokamak discharge response is always strongly related to short-term input changes but it is also affected by long-term input changes. For instance, W_{mhd} hardly changes fast (see figure 5), this property can be regarded as a short-term dependence. However, the impact of heating on stored energy W_{mhd} is cumulative. This property can be seen as long-term dependence. The dropout layer randomly sets input units to 0 with a frequency of rate (in this paper, the dropout rate is 0.1) at each step during training time, which helps prevent overfitting [81]. The final component is the fully connected layer to match the high-dimensional decoder output with the real target dimension.

When considering the specific mathematical principles of the model, the encoder hidden states h_t are computed using this formula:

$$h_t = f(W^{(h_{lstm0}\delta_{dropout0}h_{lstm1})}h_{t-1} + W^{(hx)}x_t), \tag{1}$$

where h_{lstm0}, h_{lstm1} are the hidden state of LSTM_0 and LSTM_1 in figure 2, $\delta_{dropout0}$ is the dropout rate in Dropout_0 in figure 2, W and W^{hx} represent the appropriate weights to the previously hidden state h_{t-1} and the input vector x_t. $\delta_{dropout0} \sim Bernoulli(p)$. This means

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$\delta_{dropout0}$ is equal to 1 with probability p and 0 otherwise, we let $p = 0.9$ for all experiment. Dropout_0 means that not all hidden states of LSTM_0 can be transferred to LSTM_1.

The encoder vector is the final hidden state produced from the encoder part of the model. It is calculated using the formula above. This vector aims to encapsulate the information for all input elements to help the decoder make accurate predictions. It *acts as the initial hidden state* of the decoder part of the model.

In decoder a stack of two LSTM units where each predicts an output y_t at a time step t. Each LSTM unit accepts a hidden state from the previous unit and produces and output as well as its hidden state. In the modeling of tokamak discharge, the output sequence is a collection of all time steps from the y. Any hidden states h_t is computed using the formula:

$$h_t = f(W^{(h_{lstm2}\delta_{dropout1}h_{lstm3})}h_{t-1}), \tag{2}$$

The h_{lstm2},$h_{lstm3}, \delta_{dropout1}, W$ of equation 2 have similar meaning to those in equation 1, where $\delta_{dropout1} \sim Bernoulli(0.9)$.

The process of data flowing in the decoder is similar to the encoder, but the initial state of h_{lstm2} is equal to the last state of h_{lstm1}. As formula above, we are just using the previous hidden state to compute the next one. The output y_t at time step t is computed using the formula:

$$y_t = activation(W^D h_t). \tag{3}$$

The model calculates the outputs using the hidden state at the current time step together with the respective weight W^D of the fully connected layer as shown in figure 2. The fully connected layer is used to determine the final outputs. The activation function is a linear function not a rectified linear unit (ReLU) [82, 83] or others non-linear activation functions. Basically, for regression tasks, it is customary to use the linear function as the activation function because it's differentiable and it does not limit the output. We do not use ReLU as the activation of the last layer for this regression task due to the fact that they are limited and cannot generate all numbers which are needed.

4. Training

The form of the resampling function is determined according to the time characteristics of the output signal waveforms. The output signals with the same sampling function are grouped together. The input signals are resampled using the same sampling function as the output signal set to ensure that all data points follow the same time axis. In this section, model training and data processing will be introduced in detail. The training of the model (see figure 3) can be divided into five steps as follows:

(i) Obtaining the data of 68 channels (65 channels input signals and 3 channels output signals) of the selected signals from the EAST source database (include input and output signals as shown in table 1).

(ii) Applying different resampling methods based on the time characteristics of the output signal that would be modeled.

(iii) Standardizing the data with z-scores ‡.

(iv) Feeding data into the deep learning model for training.

‡ z-score is calculated by $z = (x - \mu)/\sigma$ where μ is the mean of the population. σ is the standard deviation of the population

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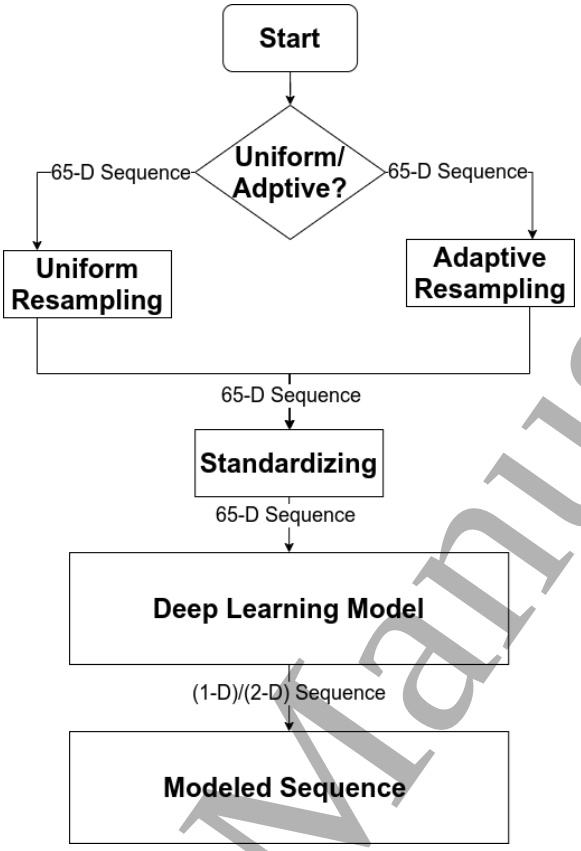

Figure 3. Workflow of training. The resampling method is determined according to the time characteristics of the output signal.

(v) Using the loss between model output and real experimental output as backing propagation metric and then update parameters for the training model.

The dataset is selected from EAST campaign 2016-2018, discharge shot number in the range #70000-80000 [74, 76, 84]. A total of 3476 normal shots are selected and divide the data into 60% training set, 20% validation set, and 20% test set. The normal shot means that no disruption occurred during this discharge, the flat-top lasts more than two seconds, and the key signals (i.e., model output signals, magnetic field, and plasma current I_p) are complete. If there is not a certain magnetic field configuration, it is impossible to constrain (or control) the plasma. And it is meaningless to model the tokamak discharge without complete I_p or model output signals. I_p is an input of the model, while the output signals are necessary to train and test the model (They will not be available when the model will be applied to model experiments.). The sampling of the signal starts from $t = 0s$ and continues to the end of the discharge (typical EAST normal discharge time is five to eight seconds).

The shuffling method is a common improved generalization technique [85] used in an entire data set. However, in the present work, this method is not used in the entire data set. In order to prevent

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data leakage caused by multiple adjacent discharge experimental shots with similar parameters, the entire data set is divided into training set, validation set and test set according to the experimental shot order. The phenomenon of adjacent discharge experimental shots with similar parameters is very common in tokamak discharge experiments. For generalization reasons, it is also necessary to shuffle shot order in the training set. For example, there are ten normal discharge shots in original data sets, with shot numbers 1-10. The training set is *shuffle*(1-6) (maybe one order is 1,4,6,5,2,3), validation set is *shuffle*(7-8), and test set is *shuffle*(9-10). Inside a single normal shot discharge, the sequence will be kept in strict time order.

After all source data is obtained, the z-scores [86] will be applied for standardization. Then all the preprocessed data will be input to the deep learning model for training. In statistics, the z-score is the number of standard deviations by which the value of a raw score (i.e., an observed value or data point) is above or below the mean value of what is being observed or measured. Raw scores above the mean have positive standard scores, while those below the mean have negative standard scores. Z-score is calculated by $z = (x - \mu)/\sigma$ where μ is the mean of the population, σ is the standard deviation of the population.

The deep learning model uses an end-to-end training that was executed on 8x Nvidia P100 GPUs with Keras [87] and TensorFlow [79] on the Centos7 operating system. The training of the deep learning model starts with kernel initializer is glorot uniform initialization [88] the recurrent initializer is orthogonal [89], bias initializer is zeros, and optimizer is Adadelta [90] for solving gradient vanish. The model trains in about twelve days, 40 epochs. Then use callbacks and checkpoints to choose the best performing model in W_{mhd} and n_e modeling best epoch is fifteen, while V_{loop} modeling best epoch is seven. In an epoch of the model training process, each sample in training set will be put into the model and updated the internal model parameters. The best epoch was selected based on the model performance on the validation set.

The training of our model is executed several times. Many of these trials are considered as failed (e.g. divergence in training, poor performance on the test set, etc) because of unsuitable hyper-parameters. In the process of training our model, multiple sets of hyperparameters were tried. It was a trial-and-error approach on different sets of hyperparameters combinations: the best hyperparameter set was chosen on the best validation set performances. Finally, the best hyper-parameters were found by and shown in table 2.

5. Results

After the deep learning model had been trained, the model can be tested on unseen data (test set). As shown in figure 4, all the data of the 65 channel signals are prepared before applying the model. During the experimental proposal stage, some input signals that can not be directly available should be reconstructed using other means (through codes or analytical models). For example, in this work, the accurate values of actual plasma current cannot be directly available, during the experimental proposal stage. But the indirectly estimated values given by the modeling program (i.e. TSC etc.) can be used as substitutes. The preprocessing steps on new data are the same as before, keeping the training set parameters (the mean μ and the standard deviation σ) for standardization. The data whose distribution lies outside of the training set, as it was scaled with the training's standardization parameters, might affect the model's generalization capabilities. All the standardized data will be fed into the trained model and get the modeling sequence of diagnostic signals. In the final step, the corresponding model should be selected according to the diagnostic signal to be modeled.

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Table 2. Hyperparameters in this model

Hyperparameter	Explanation	Best value
η	Learning rate	5×10^{-3}
γ	Adadelta decay factor	0.95
Loss function	Loss function type	Mean squared error (MSE)
Optimizer	Optimization scheme	Adadelta
Dropout	Dropout probability	0.1
Epoch	Epoch	15 and 7
dt	Time step	1ms and 0.1ms
Batch_size	Batch size	1
LSTM type	Type of LSTM	CuDNNLSTM
LSTM size	Size of the hidden state of an LSTM unit.	256
$LSTM_{kernel}$	Initializer for the kernel of LSTM weights matrix	Glorot uniform
$LSTM_{recurrent}$	Initializer for the recurrent kernel of LSTM weights matrix	Orthogonal
Dense size	Size of the Dense layer.	256
$Dense_{kernel}$	Initializer for the kernel of dense matrix	Glorot uniform
$Dense_{bias}$	Initializer for the bias vector	Zeros
$n_{encoder}$	Number of LSTMs stacked in encoder	2
$n_{decoder}$	Number of LSTMs stacked in decoder	2

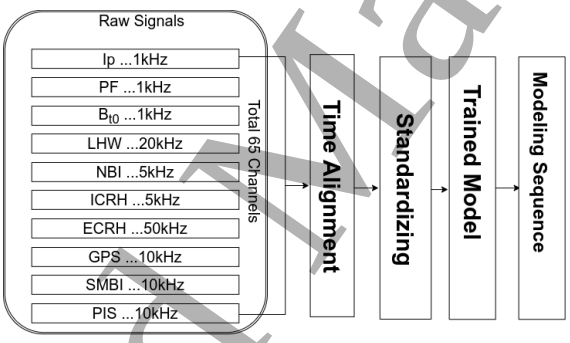

Figure 4. Using the trained model.

In this section, the results of modeling will be analyzed in detail, including representative modeling results and similarity distributions. In the present work, the similarity is a quantitative measurement of the modeling results accuracy and is defined as follows:

$$S\left(\boldsymbol{x}, \boldsymbol{y}\right) = \max\left(\frac{\Sigma(\boldsymbol{x}-\bar{\boldsymbol{x}})(\boldsymbol{y}-\bar{\boldsymbol{y}})}{\sqrt{\Sigma(\boldsymbol{x}-\bar{\boldsymbol{x}})^2\Sigma(\boldsymbol{y}-\bar{\boldsymbol{y}})^2}}, 0\right), \tag{4}$$

where $\boldsymbol{x}$ is experimental data, $\boldsymbol{y}$ is modeling result, $\bar{\boldsymbol{x}}$, $\bar{\boldsymbol{y}}$ are the means of the vector $\boldsymbol{x}$ and vector $\boldsymbol{y}$.

Two typical EAST normal discharge shots shot #77873 and #78461, are selected to check the accuracy of the model trained in this article. Figure 5(a) shows the modeling result for shot #77873, which has two LHW injections during discharge. Figure 5(b) shows the result for shot #78461, which has NBI, LHW, ICRF injection.

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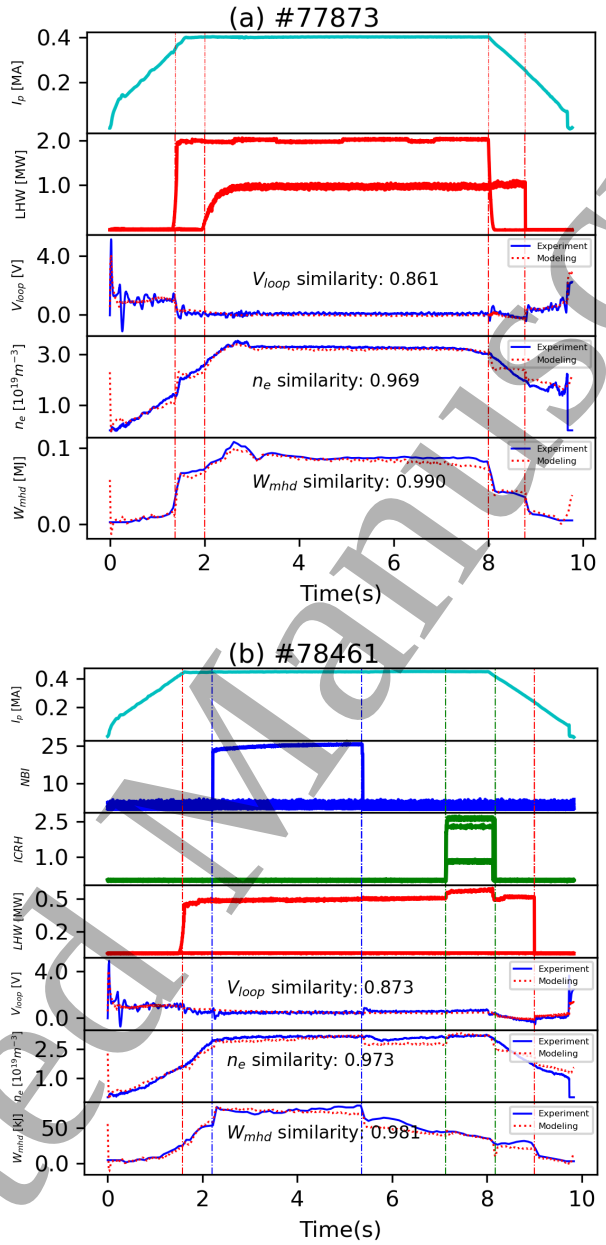

Figure 5. Comparison of modeling result and EAST experiment data, shot #77873 (a), and #78461 (b). NBI and ICRH are the raw data (these could easily be converted to Watts.) so the physical units are meaningless. NBI has eight channels, but some channels of NBI on shot #78461 have noise data.

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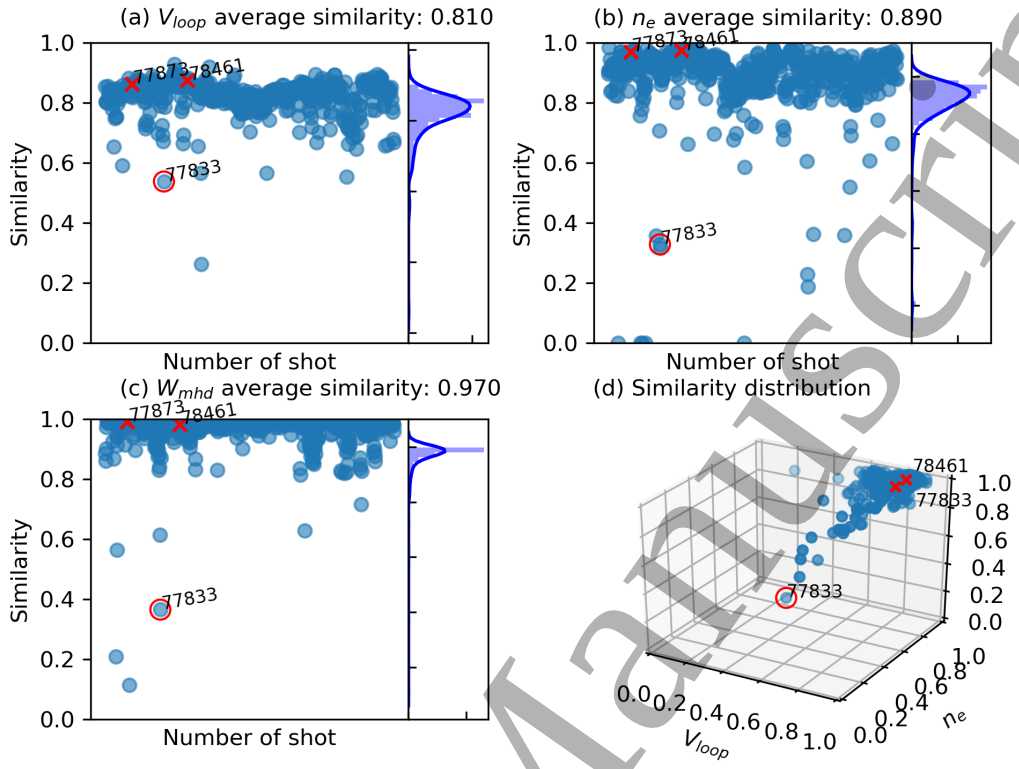

Figure 6. The similarity distribution and average similarity in the test set. The figure shows the similarity distributions of (a) n_e, (b) V_{loop} , and (c) W_{mhd}, respectively. Figure (d) is a joint scatter plot of three parameters.

Experimental data and modeling results are displayed together in figure 5. The comparison shows that they are in good agreement in most regions of discharge, from ramp-up to ramp-down. The slope of the ramp-up and the amplitude of the flat-top are accurately reproduced by the model. The vertical dash-dot lines indicate the rising and falling edges of the external auxiliary system signal and the plasma response, which show the time accuracy of the model.

Compared with experimental signals, the modeling results are more sensitive to changes of the external drives. For example, after the external drive is turned off, the experimental signal n_e continues to decrease with a fixed slope, but the modeling results show a step-down. However, it will also cause the deviation of the modeling result and the experimental data when the external drive changes rapidly. How to adjust the sensitivity of the model is still an open question.

A test data set with 695 shots were used to quantitatively evaluate the reliability of the model. The statistical results of the similarity between model results and the experimental data are shown in figure 6.

W_{mhd} is the best performance output signal, with the similarity concentrated at more than 95%. In other words, W_{mhd} can be considered to have been almost completely modeled under the normal discharge condition. The similarity distribution of n_e in the test set is concentrated above 85%. V_{loop} is the worst performing output signal. As shown in table 3, the flat-top phase is best

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	Ramp-up	Flat-top	Ramp-Down
Average similarity	0.738	0.879	0.792

Table 3. The average similarity of V_{loop} in ramp-up, flat-top, and ramp-down.

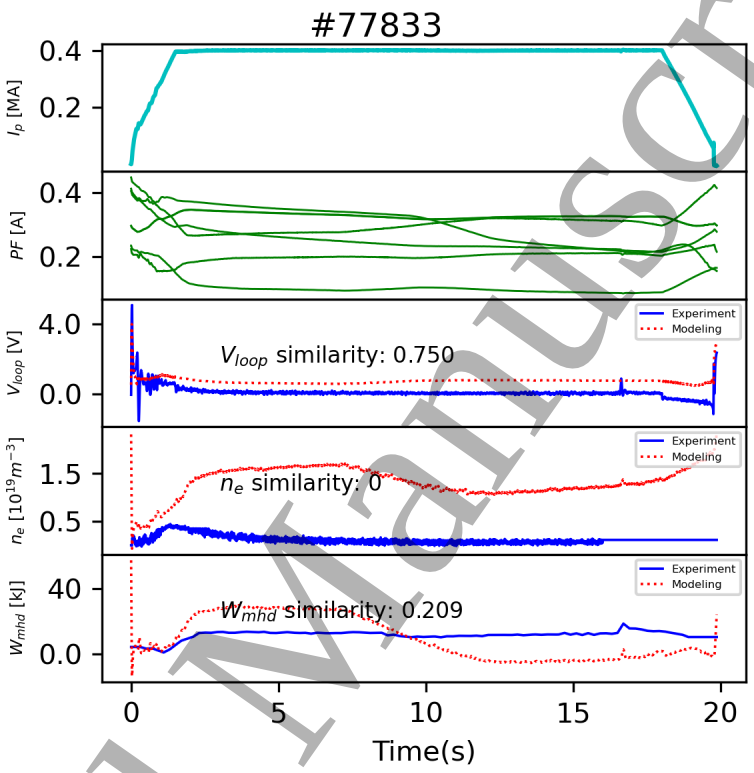

Figure 7. Shot #77833 is a classical deviation caused by abnormal device conditions. Shot #77833 is used for cleaning device.

performance phase. Many of the V_{loop} errors are due to the plasma start-up in the ramp-up phase and the plasma shutdown in the ramp-down phase.

The joint distribution of the three output signals similarities is shown in figure 6(d). Most shots are concentrated in a limited range, which reflect the consistency of the model on three target signals. It also shows that these shots belong to the same tokamak operating mode. In other words, those points far away from the center of the cluster indicate that the experiment is running in abnormal mode. We checked all deviation shots in the test set, and all deviation are in abnormal conditions. For example, shot #77833 (as shown in figure 7) is a classical deviation caused by abnormal device conditions, which is used for cleaning device.

In terms of the similarity distribution of demonstrative output signals and representative discharge modeling results, this machine learning model application in tokamak discharge modeling is promising. W_{mhd} can be regarded as almost completely reproduced under normal discharge

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shots. n_e can be successfully modeled in most areas under the normal discharge condition. In normal discharge shots, the modeling results of V_{loop} at ramp-down and flat-top phases are in good fitting with the experimental results.

6. Conclusion

In the present work, we showed the possibility of modeling the tokamak discharge process using experimental data-driven methods. A machine learning model based on the encoder-decoder was developed and trained with the EAST experimental data set. This model can use the actuators signals (i.e. NBI, ICRH, etc.) to reproduce the normal discharge process (i.e. electron density n_e, store energy W_{mhd} and loop voltage V_{loop}) without additional introducing physical models. Up to 95% similarity was achieved for W_{mhd}. Despite the recent work of discharge modeling has focused on physical-driven "Integrated Modeling", this work shows promising results for the modeling of tokamak discharge by using the data-driven methodology. This model can be easily extended to more physical quantities, and then a more comprehensive tokamak discharge model can be established.

Checking the physical goal of the experimental proposal is an important and complicated problem. This work provides a reference for the realization of physical goals under the normal discharge condition. Specifically, the model mainly checks whether an experimental proposal can be achieved under the normal discharge condition. Furthermore, if the experimental result deviates greatly from the result of our model, there may be two situations. One is that the experiment has some problems (as shown in figure 7); another one is that a new discharge mode has appeared. The reason why a new discharge mode can be found is that this model only models discharge mode that has appeared in EAST normal discharges. Of note, if other discharge modes appear in the data, this model will produce noisy output as if it were abnormal shots [91]. In general, noisy model output means wrong input or new input data not seen in the training process. For an experimental proposal, if the model gives a noisy output, the input should be carefully checked by an experienced experimenter. If the input can be confirmed to be correct then it is likely to be a new discharge mode. A limitation of the model presented in the present work is the inability to fully and accurately recognize badly preprogrammed actuators signals.

The data-driven approaches can directly construct the model without considering the physical process behind the data. This does not mean that physics is not important in the tokamak discharge modeling. The method described in the present work is only the first step in data-driven modeling. Next, an in-depth analysis of the data model is required to determine the dependence of actuators signals and diagnostic signals, and find redundant signals. Then the physical model behind the data can be established based on these results. This will be a long-term work and should be further studied and gradually improved in future work.

The present work demonstrates that data-driven methodology has the potential to provide an alternative to the conventional physical-driven approaches for tokamak discharge modeling. The present model is just a prototype. By optimizing parameters and algorithms, its performance has room for further improvement, and it may be expanded to more application scenarios. For example, the impact of model sensitivity on modeling results has been recognized. How to adjust the sensitivity of the model is still an open question. Cross-device modeling is more important for devices under design and construction such as ITER and CFETR. Introducing device configuration parameters and performing transfer learning is a feasible solution to this problem. Reproducing the time evolution of one-dimensional profiles and two-dimensional magnetic surfaces are also very

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important applications for tokamak discharge modeling.

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